Asian American Network for Cancer Awareness, Research, and Training (AANCART): Fifth Asian American Cancer Control Academy

Cancer Incidence in the Hmong in California, 1988–2000

Paul K. Mills, Ph.D., M.P.H.[1,2]
Richard C. Yang, M.P.H.[1]
Deborah Riordan, M.P.H.[1]

[1] Cancer Registry of Central California, Fresno, California.

[2] Department of Family and Community Medicine, Fresno Medical Education Program, University of California–San Francisco, San Francisco, California.

The Hmong represent a unique new Southeast Asian immigrant group to the U.S. Approximately 169,000 Hmong reside in the U.S., primarily in California, Minnesota, and Wisconsin. Previous studies of cancer in this population have indicated that Hmong experience an elevated risk of gastric, hepatic, cervical, and nasopharyngeal cancers and experience a reduced risk of breast, prostate, lung, and colorectal cancers. Approximately 65,000 Hmong live in California, where there has been a population-based cancer registry since 1988, and the authors used these data to calculate age-adjusted cancer incidence rates and to examine disease stage and tumor grade at diagnosis. Changes in rates during the period studied also were evaluated. These rates and proportions were compared with rates among the non-Hispanic white (NHW) and Asian/Pacific Islander (API) populations of California. Between 1988 and 2000, a total of 749 Hmong in California were diagnosed with invasive cancer, and the age-adjusted rate of cancer for the Hmong was 284 per 100,000 population, compared with 362.6 and 478 per 100,000 in the API and NHW populations, respectively. The age-adjusted incidence rates of cancer in the Hmong were elevated for hepatic, gastric, cervical, and nasopharyngeal cancers and for leukemia and non-Hodgkin lymphoma (NHL). Rates were lower in the Hmong for colorectal, lung, breast, and prostate cancers. For gastric cancer and lung cancer, age-adjusted rates increased between 1988 and 2000 in the Hmong, although breast cancer incidence declined. Cervical cancer incidence increased, rates of NHL were declining, and rates for colorectal cancer remained steady between 1988 and 2000. The Hmong experienced later disease stage at diagnosis than other API and generally poorer grade of disease at diagnosis. Hmong experienced lower overall invasive cancer incidence rates than API or NHW populations in California. However, they experienced higher rates of hepatic, gastric, cervical, and nasopharyngeal cancers; and, for most types of cancer, they were diagnosed in a later disease stage. ***Cancer* 2005;104(12 Suppl):2969–74.**

KEYWORDS: Hmong; cancer incidence; epidemiology; cancer registries; surveillance; Asian American Network for Cancer Awareness, Research, and Training; Asian American.

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Address for reprints: Paul K. Mills, Ph.D., M.P.H., Cancer Registry of Central California, 1320 E. Shaw Avenue, Suite 160, Fresno, CA 93710, Fax: (559) 222-8960; E-mail: paul.mills@fresno.ucsf.edu

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The Hmong are a unique new immigrant and refugee population in the U.S. They arrived in the U.S. in the last 25 years and currently number approximately 169,000 individuals residing primarily in California, Minnesota, and Wisconsin.[1]

Originally an ethnic minority in China, the Hmong fled persecution there and inhabited the mountainous regions of northern Vietnam and Laos for the last 200 years, where they engaged in subsistence, slash-and-burn farming. Because of their allegiance to the U.S.

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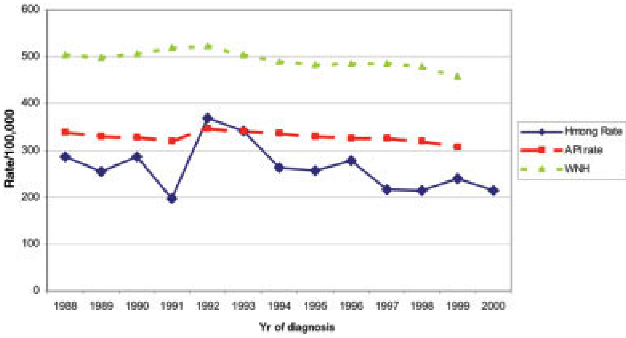

FIGURE 1. Age-adjusted invasive cancer incidence rates are illustrated for Hmong, Asian/Pacific Islander (API), and white non-Hispanic (WHN) populations in California, 1988–2000 (both genders combined).

during the Vietnam conflict, they were persecuted by the victorious Communists in Vietnam and Laos and were forced to flee to Thailand, where they lived in refugee resettlement camps. Beginning in the early 1980s, thousands of Hmong were relocated by the U.S. government to the Central Valley of California and to St. Paul, Minnesota.

Today, the Hmong are becoming more acculturated to the western lifestyle; many have been born in the U.S. and they constitute a growing minority population. However, many Hmong adhere to animist beliefs and practice Shamanism, and they may be skeptical about the benefits of western medicine.

There are approximately 65,000 Hmong residing in California, and they are characterized by a very young age structure (the median age in 1990 was 12 yrs), lower socioeconomic status, and low education levels. Statewide, approximately 44% of Hmong have less than a 9th-grade education, and only 18% have a high school diploma or equivalent degree. One-half-of all Hmong families are impoverished and are living under the Federal Poverty Level. Greater than 54% of Hmong families average $\geq$ 7 members per household.[2]

Previous cancer studies in this population in the U.S. have been based on proportionate incidence analyses in California[3] and Minnesota[4] and have revealed excess risk in the Hmong for stomach, liver, cervical, and nasopharyngeal (NPC) cancers but low risk for many western cancers, including prostate and breast cancers. In this article, we report on cancer incidence rates in the Hmong in the state of California for the years 1988–2000.

MATERIALS AND METHODS

This was a population-based study, and we used cancer registry data that were obtained through the California Cancer Registry (CCR). The CCR is a population-based cancer registry that has been monitoring the occurrence of all newly diagnosed cancers and cancer-related deaths in California since 1988. The methodology of the CCR has been described previously.[5] Patients who were diagnosed with cancer in the Hmong population between 1988 and 2000 in California were identified by examining names and other personal identifiers of all individuals who originally were classified as Hmong, Vietnamese, Laotian, Cambodian, Thai, or "other Southeast Asian" in the data base of the CCR. This review was conducted by one of the authors (R.C.Y.), who is Hmong and is familiar with Hmong names and name-giving customs. Previous studies have revealed a substantial amount of ethnic misclassification in these population groups, particularly between Hmong and Laotians.

After completing the process in which Hmong patients with cancer were identified in the CCR data base, calculation of age-specific and age-adjusted cancer incidence rates was completed using the Hmong patients as numerator data. Denominator data were derived by using U.S. Census data from the 1990 and 2000 decennial censuses and by completing linear interpolation on an age, gender, and calendar year-specific basis for the intercensal years and by extrapolating backward for the years 1988 and 1989.

Age-adjusted incidence rates were calculated with the direct method using the year 2000 U.S. population as the standard. Standard errors also were calculated for the age-adjusted rates. The age-adjusted rates were the compared with incidence rates in the Asian/Pacific Islander (API) and white non-Hispanic (NHW) populations of California for the same period. In addition to calculating rates, disease stage at diagnosis for several specific forms of cancer also was calculated for the Hmong and compared with the disease stage at diagnosis in the API population of California for the same period.

RESULTS

Between 1988 and 2000, 749 patients were diagnosed with cancer among the Hmong population of California (684 invasive cancers and 65 in situ cancers). Of these, 55.8% of the patients with cancer were female, and 44.2% were male. There was a steady increase in the number of patients diagnosed annually between 1988 (31 patients) and 2000 (59 patients). The median age at cancer diagnosis in the Hmong was 54 years. The major cancer types in the Hmong included stomach (76 patients), lung (61 patients), leukemia (55 patients), liver (48 patients), colorectal (45 patients), non-Hodgkin lymphoma (45 patients), breast (43 patients), and NPC (39 patients).

TABLE 1
Cancer Counts and Age-Adjusted Invasive Cancer Incidence Rates per 100,000 Population for Hmong and Asian/Pacific Islanders, by Gender, in California, 1988–2000: All Cancer Sites Combined

	Both genders			Males			Females		
	Hmong			Hmong			Hmong		
Yr of diagnosis	No.	Rate	API rate	No.	Rate	API rate	No.	Rage	API rate
1988	31	286.7	338.9	17	296.6	379.0	14	393.9	305.2
1989	40	255.4	330.0	25	426.2	377.7	15	157.8	291.2
1990	43	286.2	328.2	18	399.4	372.0	25	249.7	294.1
1991	32	196.4	321.0	13	237.3	368.9	19	185.4	283.2
1992	57	369.6	347.0	24	387.7	413.5	33	373.7	295.7
1993	65	342.0	342.4	25	340.8	408.9	40	367.5	292.5
1994	58	262.5	336.7	25	218.7	349.9	33	296.3	294.6
1995	58	255.5	330.6	29	281.3	383.5	29	243.8	293.1
1996	63	277.5	327.3	29	275.2	369.6	34	283.3	298.7
1997	54	217.3	327.2	24	196.7	370.6	30	239.5	298.4
1998	57	214.5	319.1	27	222.5	357.2	30	218.7	294.3
1999	67	238.9	307.6	36	269.5	339.0	31	219.5	287.2
2000	59	214.1	309.5	30	250.1	350.6	29	189.5	282.3

API: Asian/Pacific Islanders.

For both genders combined, the age-adjusted incidence rates for Hmong were lower compared with the rates for the API population of California for the same period, although the relatively small numbers of Hmong patients with cancer generate unstable incidence rates. For the period 1988–2000, the average, annual, age-adjusted incidence rate in the Hmong was 284 per 100,000, compared with 362.6 per 100,000 in the California API population and 478 per 100,000 in the California NHW population. Therefore, compared with NHW Californians during the same period, Hmong experienced a 40% lower age-adjusted incidence rate (Fig. 1).

Among males, the cancer rate for the Hmong was lower than among API for all years except for 1992, and it is on the decrease, as it is among API males in California. Among females, the trend was not as apparent, with numbers of patients and rates peaking in 1993 and 1994 and declining beginning in 1996 (Table 1).

Rates of cancer in the Hmong were much lower than in the API population for prostate, breast, lung, and colorectal cancers but were higher than in the API population for leukemia, NPC, liver cancer, and stomach cancer. Rates for uterine corpus cancer were approximately the same in Hmong women and API women, although uterine cervical cancer rates clearly were elevated in the Hmong (Table 2).

An evaluation of time trends in cancer incidence for several of the major cancer sites in the Hmong is presented in Table 3 and in Figures 2 and 3. For stomach and lung cancers, age-adjusted rates increased between 1988 and 2000 in the Hmong. The pattern was less clear for breast cancer, in which rates declined between 1988 and 2000. Cervical cancer incidence increased, whereas rates of non-Hodgkin lymphoma appeared to decline. Rates for colorectal cancer remained steady between 1988 and 2000.

Disease stage at diagnosis was evaluated for cancers that were diagnosed in the Hmong between 1988 and 2000 (Table 4). For all cancer sites combined, only 23.8% of all cancers were diagnosed at an early stage (i.e., in situ or local stage). For breast cancer, 46.5% of tumors were diagnosed at an early stage; whereas, for colorectal cancer, the rate was 20%. Only 14.6% of liver cancers were discovered early, and only 10.5% of stomach cancers were diagnosed early.

At the end of the most recent follow-up (December 2000), 33.9% of the Hmong patients with cancer remained alive. This differed by gender, however, in that 42.8% of females remained alive, but only 22.7% of males remained alive.

Finally, the rates of cancer in the leading cancer sites in Hmong males and females were compared with rates in the NHW population of California during the same period (Table 5). For hepatic, cervical, and gastric cancers; for NPC; and for leukemia, the Hmong experienced from 2-fold (for leukemia) to 35-fold (for NPC) elevations in their cancer risk compared with the NHW population.

DISCUSSION

The findings from the current study largely agree with the previous proportionate analyses conducted in Cal-

TABLE 2
Age-Adjusted Cancer Incidence Rates per 100,000 for Several Cancer Sites in Hmong, Asian/Pacific Islander, and Non-Hispanic White Population in California, 1988–2000, by Year of Diagnosis, Site, and Gender

	Hmong			
Cancer type	No.	Rate	API rate[a]	NHW rate[a]
All sites combined				
Male	322	284.0	362.6	545.5
Female	362	246.5	294.0	435.7
Liver				
Male	37	25.7	22.6	4.8
Female	11	8.8	7.7	1.8
Stomach				
Male	38	32.9	22.0	10.3
Female	36	30.0	12.7	4.3
CRC				
Male	18	13.7	52.1	62.0
Female	24	13.9	37.6	44.5
Lung				
Male	39	35.8	58.9	82.7
Female	22	17.9	27.3	57.8
NHL				
Male	31	23.6	15.3	24.4
Female	14	6.6	10.9	15.3
Leukemia				
Male	27	29.9	9.3	15.6
Female	28	17.7	6.0	9.0
Breast (female)	39	23.8	37.4	145.5
Cervix (female)	57	36.6	11.8	8.0
NPC				
Male	28	16.0	5.7	0.6
Female	11	6.9	2.1	0.2
Corpus uteri (female)	18	11.5	15.2	24.9
Prostate (male)	14	14.3	86.5	148.8

API: Asian/Pacific Islander; NHW: non-Hispanic white; CRC: colorectal cancer; NHL: non-Hodgkin lymphoma; NPC: nasopharyngeal cancer.
[a] The Asian/Pacific Islander and non-Hispanic white rates are for 1995–1999.

ifornia and Minnesota. Hmong in general have lower age-adjusted cancer incidence rates than other API populations living in California, and they have lower rates than the NHW population. However, this overall pattern masks higher rates in the Hmong for several specific forms of cancer, including cervical, gastric, and hepatic cancers and NPC. However, their rates are lower for many western forms of cancer, including lung, breast, colorectal, and prostate cancers.

In this series of Hmong patients with cancer, nearly all patients were not born in the U.S. in that 51% listed Laos as their birthplace, and 8.9% listed another Southeast Asian country. Only 5.1% reported being born in the U.S. However, information on birthplace was missing for 34.9% of the patients.

TABLE 3
Time Trends for Cancer Incidence for the Hmong in California for Several Cancer Sites, 1988–2000: Both Genders Combined

Disease type/yr of diagnosis	Rate per 100,000 population
Liver (n = 48 patients)	
1988–1991	16.9
1992–1995	19.1
1996–2000	17.7
Stomach (n = 74 patients)	
1988–1991	26.8
1992–1995	28.7
1996–2000	36.0
Lung (n = 61 patients)	
1988–1991	16.7
1992–1995	32.6
1996–2000	27.0
NHL (n = 45 patients)	
1988–1991	29.3
1992–1995	17.3
1996–2000	7.6
Leukemia (n = 55 patients)	
1988–1991	29.9
1992–1995	38.8
1996–2000	12.9
Colorectal (n = 42 patients)	
1988–1991	13.6
1992–1995	10.7
1996–2000	14.7
Breast (n = 39 patients)	
1988–1991	29.6
1992–1995	21
1996–2000	23.7
Cervix (n = 57 patients)	
1988–1991	37.8
1992–1995	48
1996–2000	30.8

NHL: non-Hodgkin lymphoma.

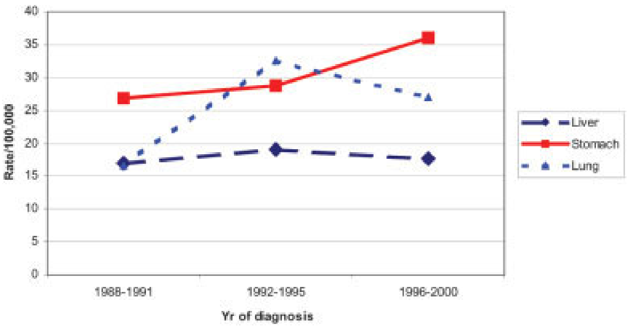

FIGURE 2. Time trends in incidence rates of liver, stomach, and lung cancers are illustrated for the California Hmong population, 1988–2000.

There was an increase of approximately 34% in the Hmong population in California during the 10 years between 1990 and 2000. However, the growth was uneven and even may have decreased in some years because of welfare-reform efforts, which caused

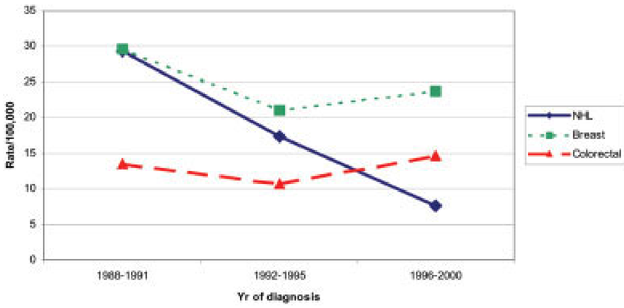

FIGURE 3. Time trends in cancer incidence in the California Hmong population, 1988–2000, are illustrated for non-Hodgkin lymphoma (NHL), breast cancer, and colorectal cancer.

some Hmong to emigrate from California in the mid-1990s, although the impact of this outflow is difficult to evaluate.

Among all API in California in 2000, the percentages of early-stage cancers were 68% for breast cancer, 69% for prostate cancer, and 37–41% for colorectal cancer. These percentages are higher compared with those observed in the Hmong, suggesting that Hmong may not avail themselves of cancer screening programs as much as other populations. There were only 14 men who had prostate cancer diagnosed in the Hmong population during 1988–2000; this may reflect the low risk for this disease, but it also may be a reflection of a low screening prevalence for this disease with prostate-specific antigen testing. Hmong are reluctant to have blood samples taken, and this may preclude their involvement in such screening programs.

In the current analysis, the median age at cancer diagnosis in the Hmong was 54 years, which contrasts with a median age at diagnosis of cancer among the API population in California of 65 years.[6] This clearly reflects differences in the age distribution of the two populations and, in particular, the extremely young age structure of the Hmong population. Among the 55 patients with leukemia observed in the Hmong between 1988 and 2000, 54.5% of patients were younger than age 20 years at diagnosis.

Similar to the Hmong, the risk of NPC also is elevated in Vietnamese immigrants to Los Angeles County[7] and in southern Chinese populations.[8] Several risk factors, including salted fish consumption and infection with Epstein–Barr virus, have been investigated. However, in the Hmong population in California, the consumption of salted fish products is not a common dietary practice.

The risk of primary hepatocellular carcinoma is associated with chronic infection with hepatitis B virus, and it is believed that approximately 80% of the liver cancers worldwide are attributable to prior infection. In California, mandatory vaccination of school children against hepatitis B virus infection has been in effect since the early 1990s; thus, the extremely high rates of liver cancer observed in the Hmong should decrease within a generation. Other risk factors, including the consumption of foods contaminated by aflatoxins and excessive alcohol consumption, need to be investigated in this population.

The elevated rates of cervical cancer in the Hmong also have been observed in other Southeast Asian immigrant populations in California, particularly among the Vietnamese.[7] A previous evaluation of cervical cancer in this population was completed[9] and identified the excess burden of cervical cancer as well as the need for Papanicolaou smear screening and outreach activities in the population.

Stomach cancer rates are elevated in the Hmong like in other Asian populations and Hispanics in California. In 1999, the age-adjusted incidence rate of stomach cancer was 14.7 per 100,000 in the API population of California and 15.9 per 100,000 in the Hispanic population.[10] The subsite distribution of this form of cancer in the Hmong shows proportionately more cancer of the gastric antrum and less cancer of the gastric cardia. *Helicobacter pylori* infection may predominate as the major risk factor for stomach cancer in the Hmong, although dietary and smoking habits also need to be evaluated.

The low rates of many western-style cancers, including breast, prostate, and colorectal cancers, reflect patterns that were observed previously in other Asian migrant groups to the U.S., particularly the Japanese and the Chinese.[11,12] However, it has been shown that these low rates increase within one generation, and continued surveillance of the Hmong is critical.

Smoking tobacco is not common in the Hmong (prevalence was estimated at approximately 17% in 1 population survey[13]), and this is reflected in the relatively low lung cancer rates in the population. However, as younger Hmong become more westernized, consumption may increase, and lung cancer rates may increase within a generation.

Limitations of the current study include the small numbers of cancers available for analysis. These relatively small numbers will result in unstable cancer rates, which should be interpreted with some caution. In addition, the accuracy of the Census data upon which the rates were based is of some concern, as is the ethnic misclassification of Hmong. Some Hmong surnames are common to Chinese surnames, such as Yang and Lee, although we have evaluated place of birth and other variables, including names of next of

TABLE 4
Percent Distribution of Stage at Diagnosis Among Solid Tumors Diagnosed in California Hmong, 1988–2001

Cancer site	No. of patients	Disease stage (%)				
		In situ	Local	Regional	Remote	Unstaged
All cancer sites combined	749	8.3	15.5	15.0	31.1	30.2
Nasopharynx	39	0.0	2.6	15.4	51.3	30.8
Stomach	76	2.6	7.9	26.3	27.6	35.5
Colorectal	45	6.7	13.3	35.6	13.3	31.1
Liver	48	0.0	14.6	4.2	35.4	45.8
Pancreas	18	0.0	0.0	11.1	50.0	38.9
Lung and bronchus	61	4.9	6.6	0.0	50.8	37.7
Breast	43	9.3	37.2	27.9	11.6	14.0
Cervix[a]	57	—	31.6	35.1	15.8	17.5
Uterine corpus	19	5.3	42.1	10.5	5.3	36.8
Prostate	14	0.0	35.7	7.1	35.7	21.4
Urinary bladder	15	40	46.7	0.0	6.7	6.7
Thyroid	22	0.0	31.8	18.2	36.4	13.6

[a] Invasive cervical cancer only.

TABLE 5
Leading Cancer Sites in California Hmong by Gender, 1988–2000, and Ratio of Hmong Rate to Non-Hispanic White Rate

Cancer site	Hmong rate		Hmong/NHW ratio	
	Males	Females	Males	Females
Liver	25.7	8.8	5.4	4.9
Cervix	—	36.6	—	4.6
Stomach	32.9	30	3.2	7.0
Lung	35.8	17.9	0.4	0.3
NPC	16.0	6.9	26.7	34.5
Leukemia	29.9	17.7	1.9	2.0

NHW: non-Hispanic whites; NPC: nasopharyngeal cancer.

kin, to minimize the number of patients who were misclassified.

The Hmong population constitutes a unique opportunity to monitor changes in cancer risk from a low-risk to a high-risk environment. Changes in lifestyle, including dietary habits, tobacco and alcohol consumption, childbearing and breastfeeding practices, and occupational factors, all contribute to cancer risk, because this population is passing through a period of dramatic change in these factors as it transitions to a more westernized life style.

REFERENCES

1. Pfeifer ME. Trends in Hmong population distribution across regions of the United States. Minneapolis, MN: Hmong Studies Resource Center. 2001.
2. U.S. Bureau of the Census. Summary file 4. Washington, DC: Public Information Office, 2003.
3. Mills PK, Yang R. Cancer incidence in the Hmong of central California (USA). *Cancer Causes Control.* 1997;8:705–712.
4. Ross J, Xie Y, Kiffmeyer W, Bushhouse S, Robinison L. Cancer in the Minnesota Hmong population. *Cancer.* 2003;97: 3076–3079.
5. California Cancer Registry, Data Standards and Assessment Unit. California cancer reporting system standards, volume 1: abstracting and coding procedures for hospitals, 5th ed. Sacramento, CA: California Department of Health Services, 1988.
6. Perkins CI, Cohen R, Young JL, et al. Cancer in California. Detailed site and histology, 1990–1994. Sacramento, CA: California Department of Health Services, Cancer Surveillance Section, 1997.
7. Ross RK, Bernstein L, Harnett NM, Boone JB. Cancer in the Vietnamese population of Los Angeles County California. *Br J Cancer.* 1991;64;185–186.
8. Yu M. Liver cancer. In: Schottenfeld D, Fraumeni JF Jr., editors. Cancer epidemiology and prevention. New York: Oxford University Press, 1996:603–618.
9. Yang R, Mills PK, Riordan D. Cervical cancer in the Hmong of California. *Am J Prev Med.* 2004;27:132–138.
10. Kwong S, Perkins CI, Morris CR, Cohen R, Allen M, Wright WE. Cancer in California: 1988–1999. Sacramento, CA: California Department of Health Services, Cancer Surveillance Section, 2001.
11. Buell P, Dunn JE Jr. Cancer mortality in Japanese Issei and Nisie of California. *Cancer.* 1965;18:656–664.
12. Thomas D, Karagas M. Migrant studies. In: Schottenfeld D, Fraumeni JF Jr., editors. Cancer epidemiology and prevention. New York: Oxford University Press, 1996:236–254.
13. Southeast Asian Tobacco Control Program. Final project report. California Health Collaborative, July 31, 2001 Fresno, CA.